

# Sam Kay AMIMA

MATHEMATICIAN · MATHEMATICS OUTREACH & COMMUNICATION · FREELANCE MUSICIAN

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## Personal profile

As a highly motivated and dedicated student with a deep passion for learning and personal growth, I consistently go above and beyond in my studies. I am recognised for my exceptional work ethic and efficient time management skills, which have enabled me to consistently meet deadlines and attain outstanding project results. I possess an unwavering aspiration to succeed and am always endeavouring to elevate myself to new heights of accomplishment and knowledge.

I am agreeable, enthusiastic and a clear communicator. A skill I am most proud of is the ability to convey complex ideas with a flexible vocabulary, tailoring conversations to a wide variety of audiences.

## Research experience

### Doctor of Philosophy (PhD) in Applied Mathematics

Durham University

Durham, UK

Oct 2025 – Present

- Working towards a PhD in solar physics under the supervision of Dr Peter Wyper.
- Developing models of various structures in the solar wind, falling under the realm of analytics in magnetohydrodynamics (MHD).

### Integrated Master of Mathematics (MMath)

Durham University

Durham, UK

Sept 2021 – July 2025

- Graduated with a Master's degree in mathematics with First Class Honours.
- Involved with the Mathematical Society.
- **Selected modules:** Advanced Quantum Theory, Fluid Mechanics, General Relativity, Geophysical & Astrophysical Fluids, Superstrings.

## Outreach programs

### City of Maths Leeds

Event day ambassador

Leeds, UK

Jan 2026 – Present

- Ambassador for the UK's first *City of Maths* project, aiming to enthrall the nation with maths and how it can be brought into everyday life.
- Events include the *Big Number Natter*, a stall at *Lark in the Park* and various ASDA shopping challenges.
- [Click here to learn more and get involved!](#)

### Chalkdust magazine

Article writer and editor

Remote

Feb 2024 – Present

- Writing and editing articles for a popular maths magazine, distributed internationally and among universities.
- Occasional cryptic crossword setter under the setter name *Seuss*.
- [Click here to read the most recent issue.](#)

### 24 Hours of Le Maths

Lead event organiser

DUMPSS

Dec 2025 – Mar 2026

- Organised the university's fifth annual 24-hour maths endurance event to raise money for *Point North*.
- Managed a team of volunteer question writers and markers; liaised directly with the charity.
- [Click here for the 2026 question paper.](#)

### Chalkboard Ultra podcast

Podcast co-host and lead editor

DU MathSoc

Oct 2023 – June 2025

- Hosting a podcast that brings light to all areas of mathematics, with conversations among PhD students and lecturers.
- [Click here for the Spotify playlist.](#)

## Skills

**Programming**    $\LaTeX$ , TikZ, Python, HTML, Markdown, Maple, Desmos, VisualPDE, ARMS, LareXd, Heliospace.

**Soft Skills**   Teamwork, problem solving, organisation, data analytics, critical thinking, report writing, presentation skills.

**Miscellaneous**   Microsoft Excel, Canva, sleight of hand and card magic.

## Research projects

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### Modelling magnetic switchbacks

Oct 2025 – Present

PhD research project

- Developing analytic models for magnetic switchbacks: rapid reversals in the Sun's ambient magnetic field that would otherwise be radial.
- Will continue this research through MHD simulations via ARMS/Lare3d and compare results to see if we can describe these phenomena.

### The Standard Model from a Symmetry Perspective

Oct 2024 – May 2025

MMath research project

- Researched the construction of the Standard Model of Particle Physics; studied gauge theory and spontaneous symmetry breaking.
- Produced a report of my research, alongside an academic poster and presentation, [available to view on my website](#).

### On Modelling Seismic Waves

Dec 2022 – Jan 2023

Mathematical Modelling module

- Conducted research into how we can simulate the effects of seismic waves to better understand the Earth's structure.
- A solo project, programming 2-dimensional partial differential equation solvers into Python.

## Work experience

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### Undergraduate tutorial leader

Durham, UK

Durham University

Oct 2025 – Present

- Assisting undergraduate students with their studies in the form of tutorials; guiding them through chosen problem sheet questions and being a mentor.
- Teaching assistant for Fluid Mechanics III; tutorial leader for Mathematical Methods II and Methods of Mathematical Physics II.

### Summer camp counsellor & Group leader

Starrucca, Pennsylvania

Island Lake Camp

Summers 2023 – 2026

- Mentor, counsellor and head of for children aged 6–16 at a sleep-away summer camp in America.
- Working as a part of the STEM, Outdoor Adventure and Rockshop departments, ensuring children were safe and engaged in their activities.

### Maths and science tutor

Remote

MyTutor.co.uk // Freelance

Jan 2022 – June 2025 // Present

- Previously tutoring in a classroom environment via video call and private tutoring with 100% return business.
- Now private tutoring on a word-of-mouth basis.

## Volunteering

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### Maths subject representative

Stephenson College

Stephenson College JCR

Jan 2025 – Present

- Point of guidance contact for students regarding mathematical sciences.
- An advisory role for those needing help with exam techniques, module choices and general academic problems.

### Jazz band president

Josephine Butler College

Butler Jazz & Butler Soul Band

Sept 2023 – Nov 2024

- Leading two bands after being the lead pianist for two years and Librarian for one year prior.
- Directing rehearsals, booking gigs, organising charity events, overseeing the executive team, managing band finances.
- Improved skills in performance, public speaking, organisation and networking with other bands.

## Interests

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**Outreach** Associate member of the Institute of Mathematics and its Applications (IMA).

**Fitness** Regularly attend the gym and enjoy running; enjoy long walks in unspoilt countryside.

**Performing Music** Regular at Durham Jazz Society's jazz jam nights @ Fabio's bar and Stag's Open Mic @ the Stag's Head.

**Music** Learning the guitar and piano since age eight, now playing mostly jazz and funk music. Avid fan of the keytar.

**References available upon request**

**Last updated 7 September 2026**